

☒ Easy (10 Questions)

Q1. Which mode is used to write to a file, erasing existing content?

- A. r
- B. w
- C. a
- D. r+

Answer: B

Explanation: w mode overwrites the file.

Q2. What does `chmod 755 file.py` do?

- A. Sets full permissions for all
- B. Sets rwx for owner, rx for others
- C. Removes all permissions
- D. Makes file hidden

Answer: B

Explanation: 755 = rwxr-xr-x.

Q3. What module is used for socket programming in Python?

- A. net
- B. comm
- C. socket

D. sys

Answer: C

Explanation: socket is built-in for networking.

Q4. What is the default port for HTTP?

A. 22

B. 21

C. 443

D. 80

Answer: D

Explanation: HTTP uses port 80.

Q5. What does filter() return?

A. List

B. Generator

C. Tuple

D. Dictionary

Answer: B

Explanation: filter() returns a filter object.

Q6. Which of the following is a Python web framework?

A. React

- B. Flask
- C. Nginx
- D. Apache

Answer: B

Explanation: Flask is a Python web framework.

Q7. What does `open("file.txt", "a")` do?

- A. Overwrites file
- B. Reads file
- C. Appends to file
- D. Deletes file

Answer: C

Explanation: a stands for append mode.

Q8. What does `recv(1024)` in a socket mean?

- A. Send data
- B. Close connection
- C. Receive up to 1024 bytes
- D. Create socket

Answer: C

Explanation: Receives 1024 bytes.

Q9. Which symbol represents execute permission in Linux?

- A. r
- B. w
- C. x
- D. e

Answer: C

Explanation: x stands for execute.

Q10. What command starts a simple Python web server on port 8000?

- A. python3 web
- B. python3 -m http.server 8000
- C. start web.py
- D. python runserver

Answer: B

Explanation: This command runs built-in web server.

📁 Medium (20 Questions)

Q11. What does the with keyword do in file handling?

- A. Creates a new file
- B. Keeps the file open
- C. Ensures proper file closure

D. Deletes file automatically

Answer: C

Explanation: Auto-closes the file after block.

Q12. Which of the following is used to read one line of a file?

A. read()

B. readline()

C. readlines()

D. readone()

Answer: B

Explanation: readline() returns a single line.

Q13. Which Python function sets file permissions?

A. os.permissions()

B. file.set()

C. os.chmod()

D. sys.access()

Answer: C

Explanation: os.chmod() changes file mode.

Q14. What is true about AF_INET?

A. Used for Bluetooth

- B. For Unix domain sockets
- C. Used for IPv4
- D. Used for file access

Answer: C

Explanation: AF_INET = IPv4.

Q15. In socket programming, what does bind() do?

- A. Closes the socket
- B. Attaches IP and port
- C. Sends data
- D. Listens for connection

Answer: B

Explanation: bind() assigns address and port.

Q16. What does listen() do in server socket?

- A. Starts accepting connections
- B. Sends data
- C. Creates a new port
- D. None

Answer: A

Explanation: Server waits for clients.

Q17. What's the purpose of lambda in Python?

- A. Create class
- B. Create anonymous function
- C. Define library
- D. Return integer

Answer: B

Explanation: lambda defines one-line functions.

Q18. Which of these are functional programming tools?

- A. map, reduce, filter
- B. for, while
- C. if, else
- D. try, except

Answer: A

Explanation: These are higher-order functions.

Q19. Which port does HTTPS use?

- A. 21
- B. 80
- C. 443
- D. 25

Answer: C

Explanation: HTTPS uses port 443.

Q20. What is true about web servers?

- A. They run client-side scripts
- B. Serve static and dynamic content
- C. Only store files
- D. Are databases

Answer: B

Explanation: Web servers handle content delivery.

Q21. Which framework is full-stack?

- A. Flask
- B. Django
- C. Requests
- D. urllib

Answer: B

Explanation: Django includes ORM, routing, templates, etc.

Q22. What is client-side scripting mainly used for?

- A. Server-side code
- B. Interactivity
- C. Running Python code
- D. Running Flask

Answer: B

Explanation: Adds dynamic behavior on browser.

Q23. What is the extension for Python shared libraries?

- A. .so
- B. .dll
- C. .exe
- D. .py

Answer: A

Explanation: Linux shared objects use .so.

Q24. What is the use of `accept()` in server sockets?

- A. Sends data
- B. Receives data
- C. Accepts incoming client connection
- D. Initializes socket

Answer: C

Explanation: Accepts new client connection.

Q25. What does `os.listdir()` return?

- A. Permissions
- B. List of processes
- C. List of directory contents
- D. Current directory

Answer: C

Explanation: Shows files in directory.

Q26. Which protocol does socket use by default?

- A. HTTP
- B. FTP
- C. TCP
- D. UDP

Answer: C

Explanation: SOCK_STREAM = TCP.

Q27. What is Flask?

- A. Web browser
- B. Python web framework
- C. Text editor
- D. Web server

Answer: B

Explanation: Flask helps build web apps in Python.

Q28. Which Python command installs third-party libraries?

- A. install library
- B. py install
- C. pip install
- D. python install

Answer: C

Explanation: pip is Python package manager.

Q29. What is the role of a client in networking?

- A. Provide services
- B. Request services
- C. Host website
- D. Run database

Answer: B

Explanation: Clients send requests to servers.

Q30. What will `map(lambda x: x**2, [1,2])` return?

- A. [2, 4]
- B. [1, 4]
- C. [1, 2]
- D. None

Answer: B

Explanation: Squares each element.

☒ **Hard (10 Questions)**

Q31. What does this socket code do?

python

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```
s = socket.socket() s.connect(("localhost", 12345)) s.sendall(b"hello")
```

- A. Starts server
- B. Listens for clients
- C. Sends "hello" to server
- D. None

Answer: C

Explanation: Connects and sends data to server.

Q32. Which file permission code gives full control to owner only?

- A. 777
- B. 700
- C. 755
- D. 644

Answer: B

Explanation: 700 = rwx-----.

Q33. In functional programming, which of these are stateless operations?

- A. map()
- B. filter()
- C. reduce()
- D. All

Answer: D

Explanation: All operate without modifying external state.

Q34. What's the output of this?

```
python
```

```
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```

```
with open("f.txt", "w") as f: f.write("abc\n123")
```

A. Nothing

- B. File with 2 lines
- C. File with one line
- D. Error

Answer: B

Explanation: \n adds a new line.

Q35. Which Flask decorator handles routing?

- A. @url
- B. @route
- C. @app.route()
- D. @flask.route

Answer: C

Explanation: Maps function to URL.

Q36. What is socket backlog in listen(n)?

- A. Number of simultaneous data packets
- B. Max queued connections
- C. Number of retries
- D. Max bandwidth

Answer: B

Explanation: Defines queue size for client connections.

Q37. What is the purpose of seek() in files?

- A. Write data
- B. Read from current position
- C. Change file pointer location
- D. Delete file

Answer: C

Explanation: Moves cursor to new position.

Q38. Which of the following is NOT a Python standard library?

- A. math
- B. os
- C. numpy
- D. sys

Answer: C

Explanation: NumPy is third-party.

Q39. What is the purpose of the request object in Flask?

- A. Handle HTML
- B. Serve static files
- C. Access client data (GET/POST)
- D. Redirect URLs

Answer: C

Explanation: request captures incoming data.

Q40. What does this command do: `chmod +x script.py`?

- A. Deletes file
- B. Grants write permission
- C. Makes file executable
- D. Hides the file

Answer: C

Explanation: Adds execute permission.